# 📧 Email Spam Detector

Multinomial Naive Bayes · TF-IDF (5000 features) · Dataset: local file

| Accuracy | Precision | Recall | F1 Score | ROC AUC |
|---|---|---|---|---|
| 98.6% | 97.8% | 91.3% | 94.4% | 0.991 |

## Try it — check any message

Paste or type a message below

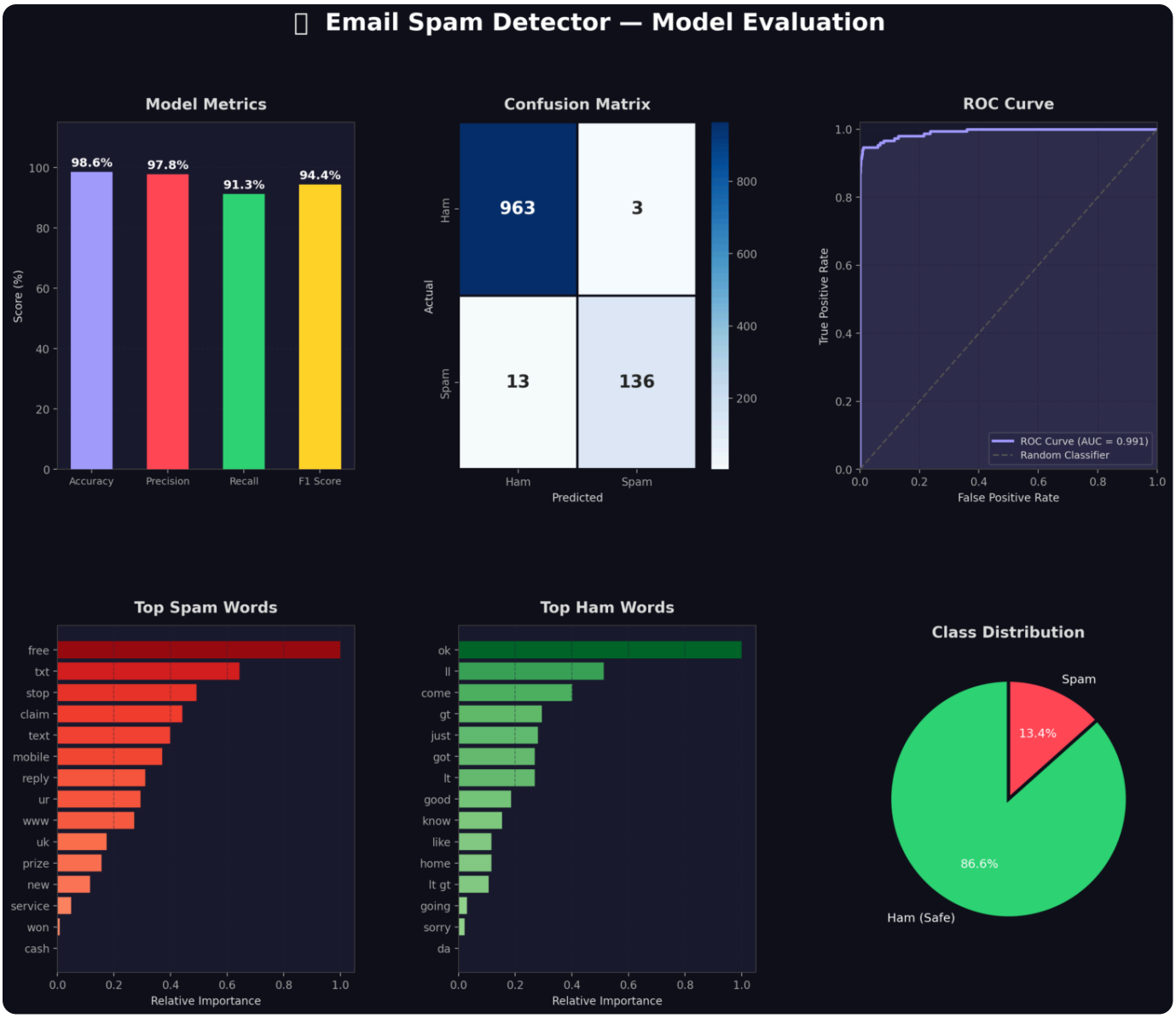
congratulations you won a free prize

Check Message

🚨 SPAM

This message is 98.6% likely to be spam

## Model Evaluation Charts


Email Spam Detector — Model Evaluation
Model Metrics
Score (%)
98.6%
97.8%
91.3%
94.4%
Accuracy
Precision
Recall
F1 Score
Confusion Matrix
Actual
Ham
Spam
963
3
13
136
Predicted
ROC Curve
True Positive Rate
False Positive Rate
ROC Curve (AUC = 0.991)
Random Classifier
Top Spam Words
free
txt
stop
claim
text
mobile
reply
ur
www
uk
prize
new
service
won
cash
Relative Importance
Top Ham Words
ok
ll
come
gt
just
got
lt
good
know
like
home
lt gt
going
sorry
da
Relative Importance
Class Distribution
Spam
13.4%
86.6%
Ham (Safe)